```
> restart;#Flux Photon force Casmir aprox;
> de := 1/c^2·H(r)·diff(u(t), t, t) + u(t)·diff(H(r), r, r) = 0;
```

$$de := \frac{H(r)\left(\frac{d^2}{dt^2}\,u(t)\right)}{c^2} + u(t)\left(\frac{d^2}{dr^2}\,H(r)\right) = 0 \tag{1}$$

```
> dsolve(de, u(t));
```

$$u(t) = _C1 \sin\left(\frac{c\sqrt{\frac{d^2}{dr^2}\,H(r)}\;t}{\sqrt{H(r)}}\right) + _C2 \cos\left(\frac{c\sqrt{\frac{d^2}{dr^2}\,H(r)}\;t}{\sqrt{H(r)}}\right) \tag{2}$$

```
>  eq1 := r·diff(H3(r), r, r) + diff(H3(r), r) + (ν^2·r - b^2·r^2/R)·H3(r) = 0;
```

$$eq1 := r\left(\frac{d^2}{dr^2}\,H3(r)\right) + \frac{d}{dr}\,H3(r) + \left(\nu^2 r - \frac{b^2 r^2}{R}\right) H3(r) = 0 \tag{3}$$

```
> eq2 := (r·diff(H3(r), r, r) + diff(H3(r), r) + (ν^2·r - b^2·r^2/R)·H3(r))/r = 0;
```

$$eq2 := \frac{r\left(\frac{d^2}{dr^2}\,H3(r)\right) + \frac{d}{dr}\,H3(r) + \left(\nu^2 r - \frac{b^2 r^2}{R}\right) H3(r)}{r} = 0 \tag{4}$$

```
> # Thus with R:=3.9228·10^22 m  we can  ignore the therm -b^2·r/R for
  1/R < r < R;
```

$$\frac{1}{R} < r < R \tag{5}$$

```
> 1/r diff((r·diff H(r), r), r) + ν^2· H(r) = 0 ;
```

$$\frac{2\,diff\left(\frac{d}{dr}\,H(r)\right) + r\,diff\left(\frac{d^2}{dr^2}\,H(r)\right)}{r} + \nu^2 H(r) = 0 \tag{6}$$

```
> de1 := 1/r·diff(r·diff(H(r), r), r) + ν^2·H(r);
```

$$de1 := \frac{\frac{d}{dr}\,H(r) + r\left(\frac{d^2}{dr^2}\,H(r)\right)}{r} + \nu^2 H(r) \tag{7}$$

```
> bc := H(R) = 0, D(H)(1/R) = R ;
```

$$bc := H(R) = 0, \mathrm{D}(H)\left(\frac{1}{R}\right) = R \tag{8}$$

```
> dsol := dsolve({de1, bc}, H(r));
```

(9)

$$dsol := H(r) = \frac{R\,\mathrm{BesselY}(0,\nu R)\,\mathrm{BesselJ}(0,\nu r)}{\nu\left(\mathrm{BesselY}\left(1,\frac{\nu}{R}\right)\mathrm{BesselJ}(0,\nu R)-\mathrm{BesselJ}\left(1,\frac{\nu}{R}\right)\mathrm{BesselY}(0,\nu R)\right)} - \frac{\mathrm{BesselJ}(0,\nu R)\,R\,\mathrm{BesselY}(0,\nu r)}{\nu\left(\mathrm{BesselY}\left(1,\frac{\nu}{R}\right)\mathrm{BesselJ}(0,\nu R)-\mathrm{BesselJ}\left(1,\frac{\nu}{R}\right)\mathrm{BesselY}(0,\nu R)\right)} \tag{9}$$

```
> H(r) :=
  R*BesselY(0,nu*R)*BesselJ(0,nu*r)/(nu*(BesselY(1,nu/R)*BesselJ(0,nu*R)-BesselY(0,nu*R)*BesselJ(1,nu/R)))
  -(BesselJ(0,nu*R)*R*BesselY(0,nu*r))/(nu*(BesselY(1,nu/R)*BesselJ(0,nu*R)-BesselY(0,nu*R)*BesselJ(1,nu/R)));
```

$$H := r \mapsto \frac{R\cdot\mathrm{BesselY}(0,\nu\cdot R)\cdot\mathrm{BesselJ}(0,\nu\cdot r)}{\nu\cdot\left(\mathrm{BesselY}\left(1,\frac{\nu}{R}\right)\cdot\mathrm{BesselJ}(0,\nu\cdot R)-\mathrm{BesselJ}\left(1,\frac{\nu}{R}\right)\cdot\mathrm{BesselY}(0,\nu\cdot R)\right)} - \frac{\mathrm{BesselJ}(0,\nu\cdot R)\cdot R\cdot\mathrm{BesselY}(0,\nu\cdot r)}{\nu\cdot\left(\mathrm{BesselY}\left(1,\frac{\nu}{R}\right)\cdot\mathrm{BesselJ}(0,\nu\cdot R)-\mathrm{BesselJ}\left(1,\frac{\nu}{R}\right)\cdot\mathrm{BesselY}(0,\nu\cdot R)\right)} \tag{10}$$

```
> u(t) := C1*sin(c*sqrt(diff(H(r),r$2))*t/sqrt(H(r)))+C2*cos(c*sqrt(diff(H(r),r$2))*t/sqrt(H(r)));
```

$$u := t \mapsto C1\cdot\sin\left(\frac{c\cdot\sqrt{\frac{d^2}{dr^2}H(r)}\cdot t}{\sqrt{H(r)}}\right)+C2\cdot\cos\left(\frac{c\cdot\sqrt{\frac{d^2}{dr^2}H(r)}\cdot t}{\sqrt{H(r)}}\right) \tag{11}$$

```
> Hphoto := subs(H(r)
  = R*BesselY(0,nu*R)*BesselJ(0,nu*r)/(nu*(BesselY(1,nu/R)*BesselJ(0,nu*R)-BesselY(0,nu*R)*BesselJ(1,nu/R)))
  -(BesselJ(0,nu*R)*R*BesselY(0,nu*r))/(nu*(BesselY(1,nu/R)*BesselJ(0,nu*R)-BesselY(0,nu*R)*BesselJ(1,nu/R))), H(r)*u(t)/r);
  # Because we have Effective Wave Function
```

$$Hphoto := \frac{1}{r}\left(\left(\frac{R\,\mathrm{BesselY}(0,\nu R)\,\mathrm{BesselJ}(0,\nu r)}{\nu\left(\mathrm{BesselY}\left(1,\frac{\nu}{R}\right)\mathrm{BesselJ}(0,\nu R)-\mathrm{BesselJ}\left(1,\frac{\nu}{R}\right)\mathrm{BesselY}(0,\nu R)\right)} \right.\right. \tag{12}$$

$$\left. - \frac{\mathrm{BesselJ}(0,\nu R)\,R\,\mathrm{BesselY}(0,\nu r)}{\nu\left(\mathrm{BesselY}\left(1,\frac{\nu}{R}\right)\mathrm{BesselJ}(0,\nu R)-\mathrm{BesselJ}\left(1,\frac{\nu}{R}\right)\mathrm{BesselY}(0,\nu R)\right)} \right)$$

$$\left(C1\,\sin\left(\left(c \left(\right.\right.\right.\right.$$

$$\left. - \frac{R\,\mathrm{BesselY}(0,\nu R)\left(\mathrm{BesselJ}(0,\nu r)-\frac{\mathrm{BesselJ}(1,\nu r)}{\nu r}\right)\nu}{\mathrm{BesselY}\left(1,\frac{\nu}{R}\right)\mathrm{BesselJ}(0,\nu R)-\mathrm{BesselJ}\left(1,\frac{\nu}{R}\right)\mathrm{BesselY}(0,\nu R)} \right.$$

$$\left. \left. \left. + \frac{\mathrm{BesselJ}(0,\nu R)\,R\left(\mathrm{BesselY}(0,\nu r)-\frac{\mathrm{BesselY}(1,\nu r)}{\nu r}\right)\nu}{\mathrm{BesselY}\left(1,\frac{\nu}{R}\right)\mathrm{BesselJ}(0,\nu R)-\mathrm{BesselJ}\left(1,\frac{\nu}{R}\right)\mathrm{BesselY}(0,\nu R)} \right)^{1/2} t \right) \Big/$$

$$\left(\frac{R\,\mathrm{BesselY}(0,\nu R)\,\mathrm{BesselJ}(0,\nu r)}{\nu\left(\mathrm{BesselY}\left(1,\frac{\nu}{R}\right)\mathrm{BesselJ}(0,\nu R)-\mathrm{BesselJ}\left(1,\frac{\nu}{R}\right)\mathrm{BesselY}(0,\nu R)\right)} \right.$$

$$\left. - \frac{\mathrm{BesselJ}(0,\nu R)\,R\,\mathrm{BesselY}(0,\nu r)}{\nu\left(\mathrm{BesselY}\left(1,\frac{\nu}{R}\right)\mathrm{BesselJ}(0,\nu R)-\mathrm{BesselJ}\left(1,\frac{\nu}{R}\right)\mathrm{BesselY}(0,\nu R)\right)} \right)^{1/2}$$

$$+ C2\cos\left(\left(\left(c\left(\right.\right.\right.\right.$$

$$\left. - \frac{R\,\mathrm{BesselY}(0,\nu R)\left(\mathrm{BesselJ}(0,\nu r)-\frac{\mathrm{BesselJ}(1,\nu r)}{\nu r}\right)\nu}{\mathrm{BesselY}\left(1,\frac{\nu}{R}\right)\mathrm{BesselJ}(0,\nu R)-\mathrm{BesselJ}\left(1,\frac{\nu}{R}\right)\mathrm{BesselY}(0,\nu R)} \right.$$

$$\left. + \frac{\mathrm{BesselJ}(0,\nu R)\,R\left(\mathrm{BesselY}(0,\nu r)-\frac{\mathrm{BesselY}(1,\nu r)}{\nu r}\right)\nu}{\mathrm{BesselY}\left(1,\frac{\nu}{R}\right)\mathrm{BesselJ}(0,\nu R)-\mathrm{BesselJ}\left(1,\frac{\nu}{R}\right)\mathrm{BesselY}(0,\nu R)} \right)^{1/2} t \Bigg) \Bigg/$$

$$\left(\frac{R\,\mathrm{BesselY}(0,\nu R)\,\mathrm{BesselJ}(0,\nu r)}{\nu\left(\mathrm{BesselY}\left(1,\frac{\nu}{R}\right)\mathrm{BesselJ}(0,\nu R)-\mathrm{BesselJ}\left(1,\frac{\nu}{R}\right)\mathrm{BesselY}(0,\nu R)\right)} \right.$$

$$\left. \left. - \frac{\mathrm{BesselJ}(0,\nu R)\,R\,\mathrm{BesselY}(0,\nu r)}{\nu\left(\mathrm{BesselY}\left(1,\frac{\nu}{R}\right)\mathrm{BesselJ}(0,\nu R)-\mathrm{BesselJ}\left(1,\frac{\nu}{R}\right)\mathrm{BesselY}(0,\nu R)\right)} \right)^{1/2} \right) \right)$$

```
> FPhoto0 := subs( t = 0, Hphoto / (2·Pi·r·l) );
```

$FPhoto0 :=$ **(13)**

$$\frac{1}{2\,r^2\,\pi\,l}\left(\left(\frac{R\,\mathrm{BesselY}(0,\nu\,R)\,\mathrm{BesselJ}(0,\nu\,r)}{\nu\left(\mathrm{BesselY}\left(1,\frac{\nu}{R}\right)\mathrm{BesselJ}(0,\nu\,R)-\mathrm{BesselJ}\left(1,\frac{\nu}{R}\right)\mathrm{BesselY}(0,\nu\,R)\right)}\right.\right.$$
$$\left.-\frac{\mathrm{BesselJ}(0,\nu\,R)\,R\,\mathrm{BesselY}(0,\nu\,r)}{\nu\left(\mathrm{BesselY}\left(1,\frac{\nu}{R}\right)\mathrm{BesselJ}(0,\nu\,R)-\mathrm{BesselJ}\left(1,\frac{\nu}{R}\right)\mathrm{BesselY}(0,\nu\,R)\right)}\right)(C1\sin(0)$$
$$\left.+C2\cos(0))\right)$$

```
> solve(FPhoto = 0);
```

$$0 \tag{14}$$

```
> FPhoto := subs(t = 1, C1 = 1, C2 = 1, k = 1, Hphoto/(2·Pi·r·l));
```

$$FPhoto := \tag{15}$$

$$\frac{1}{2\,r^2\,\pi\,l}\left(\left(\frac{R\,\mathrm{BesselY}(0,\nu\,R)\,\mathrm{BesselJ}(0,\nu\,r)}{\nu\left(\mathrm{BesselY}\left(1,\frac{\nu}{R}\right)\mathrm{BesselJ}(0,\nu\,R)-\mathrm{BesselJ}\left(1,\frac{\nu}{R}\right)\mathrm{BesselY}(0,\nu\,R)\right)}\right.\right.$$
$$\left.-\frac{\mathrm{BesselJ}(0,\nu\,R)\,R\,\mathrm{BesselY}(0,\nu\,r)}{\nu\left(\mathrm{BesselY}\left(1,\frac{\nu}{R}\right)\mathrm{BesselJ}(0,\nu\,R)-\mathrm{BesselJ}\left(1,\frac{\nu}{R}\right)\mathrm{BesselY}(0,\nu\,R)\right)}\right)$$
$$\left(\sin\left(\left(\left(c\left(-\frac{R\,\mathrm{BesselY}(0,\nu\,R)\left(\mathrm{BesselJ}(0,\nu\,r)-\frac{\mathrm{BesselJ}(1,\nu\,r)}{\nu\,r}\right)\nu}{\mathrm{BesselY}\left(1,\frac{\nu}{R}\right)\mathrm{BesselJ}(0,\nu\,R)-\mathrm{BesselJ}\left(1,\frac{\nu}{R}\right)\mathrm{BesselY}(0,\nu\,R)}\right.\right.\right.\right.$$

$$+\frac{\mathrm{BesselJ}(0,\nu R)\,R\left(\mathrm{BesselY}(0,\nu r)-\frac{\mathrm{BesselY}(1,\nu r)}{\nu r}\right)\nu}{\mathrm{BesselY}\left(1,\frac{\nu}{R}\right)\mathrm{BesselJ}(0,\nu R)-\mathrm{BesselJ}\left(1,\frac{\nu}{R}\right)\mathrm{BesselY}(0,\nu R)}\Bigg)^{1/2}\Bigg)\Bigg/$$

$$\left(\frac{R\,\mathrm{BesselY}(0,\nu R)\,\mathrm{BesselJ}(0,\nu r)}{\nu\left(\mathrm{BesselY}\left(1,\frac{\nu}{R}\right)\mathrm{BesselJ}(0,\nu R)-\mathrm{BesselJ}\left(1,\frac{\nu}{R}\right)\mathrm{BesselY}(0,\nu R)\right)}\right.$$

$$\left.-\frac{\mathrm{BesselJ}(0,\nu R)\,R\,\mathrm{BesselY}(0,\nu r)}{\nu\left(\mathrm{BesselY}\left(1,\frac{\nu}{R}\right)\mathrm{BesselJ}(0,\nu R)-\mathrm{BesselJ}\left(1,\frac{\nu}{R}\right)\mathrm{BesselY}(0,\nu R)\right)}\right)^{1/2}\Bigg)$$

$$+\cos\Bigg(\Bigg(\Bigg(c\left(\right.\right.\right.$$

$$-\frac{R\,\mathrm{BesselY}(0,\nu R)\left(\mathrm{BesselJ}(0,\nu r)-\frac{\mathrm{BesselJ}(1,\nu r)}{\nu r}\right)\nu}{\mathrm{BesselY}\left(1,\frac{\nu}{R}\right)\mathrm{BesselJ}(0,\nu R)-\mathrm{BesselJ}\left(1,\frac{\nu}{R}\right)\mathrm{BesselY}(0,\nu R)}$$

$$+\frac{\mathrm{BesselJ}(0,\nu R)\,R\left(\mathrm{BesselY}(0,\nu r)-\frac{\mathrm{BesselY}(1,\nu r)}{\nu r}\right)\nu}{\mathrm{BesselY}\left(1,\frac{\nu}{R}\right)\mathrm{BesselJ}(0,\nu R)-\mathrm{BesselJ}\left(1,\frac{\nu}{R}\right)\mathrm{BesselY}(0,\nu R)}\Bigg)^{1/2}\Bigg)\Bigg/$$

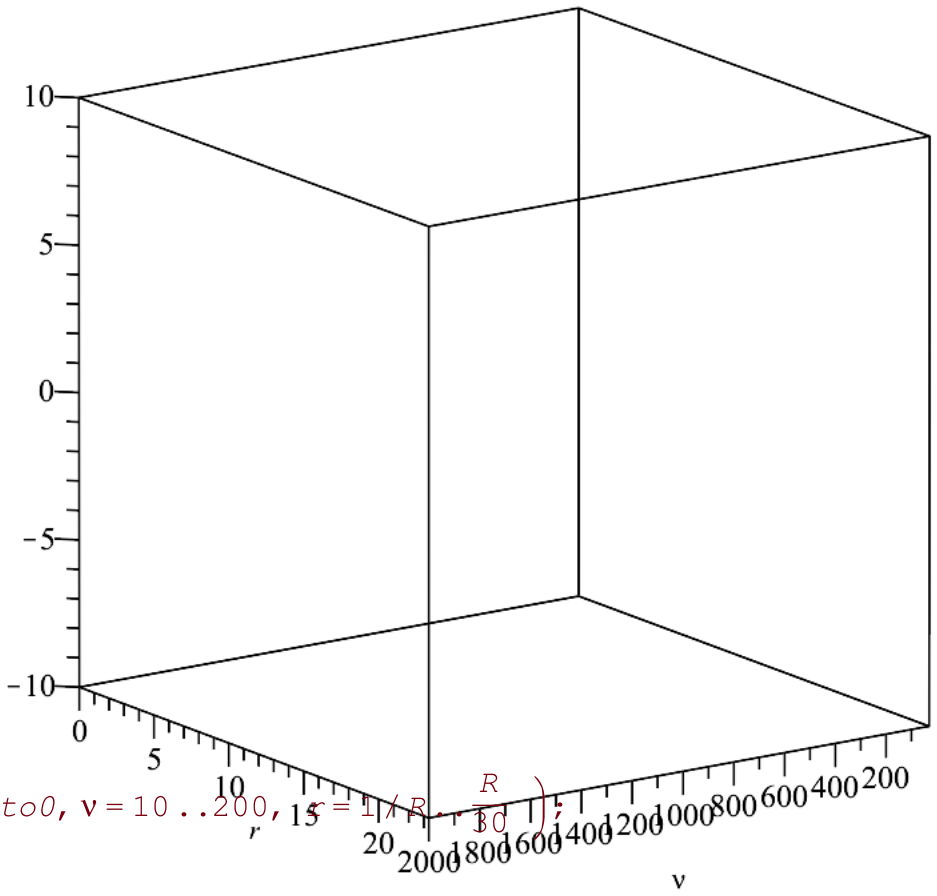
$$\left(\frac{R\,\mathrm{BesselY}(0,\nu R)\,\mathrm{BesselJ}(0,\nu r)}{\nu\left(\mathrm{BesselY}\left(1,\frac{\nu}{R}\right)\mathrm{BesselJ}(0,\nu R)-\mathrm{BesselJ}\left(1,\frac{\nu}{R}\right)\mathrm{BesselY}(0,\nu R)\right)} - \frac{\mathrm{BesselJ}(0,\nu R)\,R\,\mathrm{BesselY}(0,\nu r)}{\nu\left(\mathrm{BesselY}\left(1,\frac{\nu}{R}\right)\mathrm{BesselJ}(0,\nu R)-\mathrm{BesselJ}\left(1,\frac{\nu}{R}\right)\mathrm{BesselY}(0,\nu R)\right)} \right)^{1/2} \Bigg) \Bigg)$$

```
> R := 7·10^2;  C1 := 1; C2 := 1; c := 3000; l := 8;
```

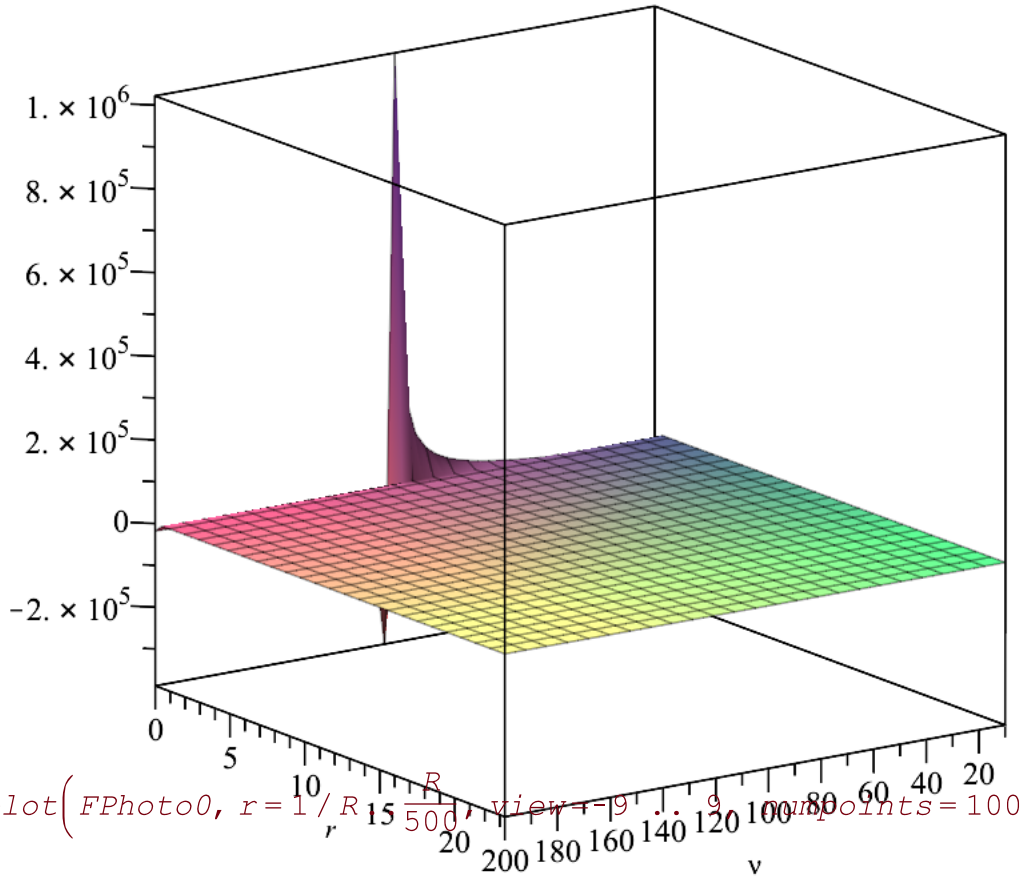
$$R := 700$$
$$C1 := 1$$
$$C2 := 1$$
$$c := 3000$$
$$l := 8 \qquad \textbf{(16)}$$

```
> plot3d( FPto, ν = 30..2000, r = 1/R..R/30 );
```

Warning, expecting only range variables [nu, r] in expression FPto to be plotted but found name FPto

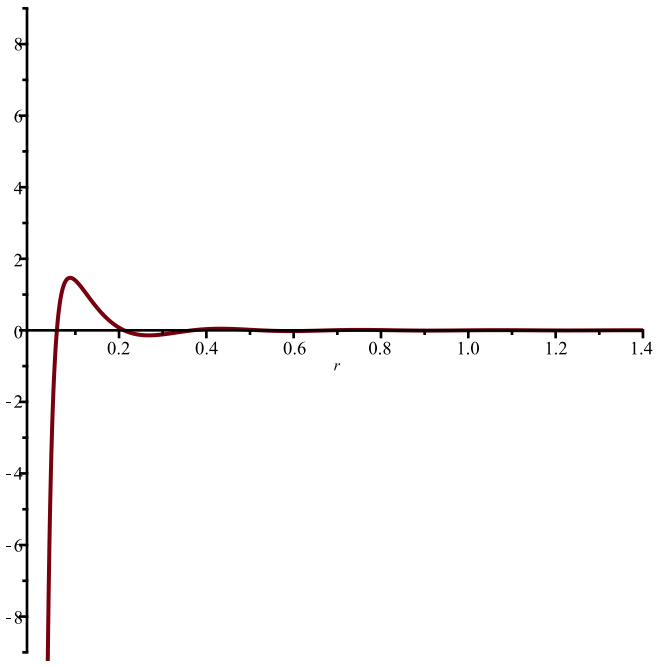



```
> plot3d( FPhoto0, ν = 10..200, r = 1/R..R/30 );
```



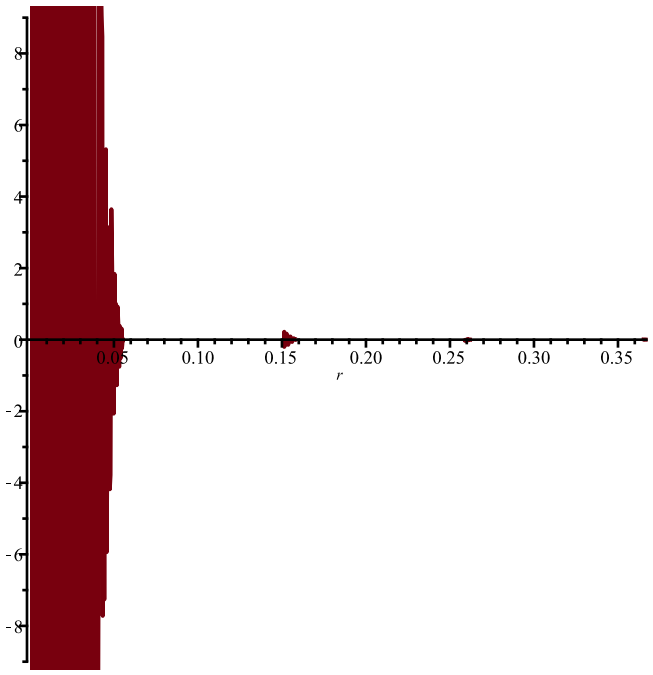

```
> ν := 20; plot( FPhoto0, r = 1 / R .. 1/500, view = -9 .. 9, numpoints = 1000 );
```

ν := 20



```
> ν := 30; plot(FPhoto, r = 1 / R .. R/1500, view = -9 .. 9, numpoints
  = 1000);
```

$$\nu := 30$$




```
>
```